

Research on Smart Navigation System Based on AR Technology

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ABSTRACT

The system uses the Unity3D engine to develop the Android app, and develop AR technology modules with Vuforia toolkit, integrating geographic information service technology and panorama technology. We combined the two tracking and registration methods which are based on sensors and natural images features, to implement a tourism navigation and AR introduction system based on Kulangsu, a famous scenic spot in Xiamen. The system is mainly divided into two modules, the route navigation module and the scenic spots guide module.

Keywords: AR technology, Navigation System, Geographic information technology

1. INTRODUCTION

Augmented reality is a new technology developed on the basis of virtual reality. It combines computer graphics (CG), Space positioning technique, visualization technology and interactive technology. It enhances the real environment by superimposing computer-generated virtual objects, scenes or system prompts into real scenes, then it will provide users with a new sensory experience. In recent years, the hardware of smart phone has developed rapidly. It is an ideal platform to develop augmented reality technology because of the advantages of fast processor, multiple sensors, GPS, high-pixel built-in camera and large storage space. At home and abroad, there have been some attempts to introduce augmented reality technology into the outdoor and indoor navigation fields, and to enhance users' perception of enhancing information diversification

The combination of geolocation navigation technology and augmented reality technology is a major trend in current research: when augmented reality technology is applied to navigation, compared with the map and compass instructions, the walking route is marked directly on the screen through a three-dimensional model or icon, which is more intuitive and convenient, effectively guiding the user to the target location, avoiding problems such as not being able to read the map and distinguish the intersection.

This paper is about a development based on Android system, we used Unity3D engine to develop the app, used the Vuforia toolkit to develop augmented reality technology modules, integrated geographic information service technology and panorama technology. We combined the two tracking and registration methods which are based on sensors and natural images features, to implement a tourism navigation and AR introduction system based on Kulangsu, a famous scenic spot in Xiamen. The system is mainly divided into two modules, the route navigation module and the scenic spots guide module. The navigation module realizes real-time navigation by integrating the three-dimensional virtual tour guide into the real world scene, so that the user is more intuitive in the process of using the navigation, and can also make the user feel immersive. The scenic spots guide module can receive relevant information of the scenic spots through the AR application on the mobile phone, view the 3D virtual scenic spots model, and perform simple interaction.

2. RELATED WORK

2.1 Geographic information technology

Geographic information technology, or "3S" technology, includes geographic information systems, remote sensing, global positioning systems and digital Earth technology. GIS collects, stores, manages, calculates, analyzes, displays and describes relevant geographic data in the whole or part of the Earth's surface space, supported by computer hardware and software systems, and is used to analyze and solve various phenomena in a certain geographic area and to solve various problems such as complicated planning, strategy and management. GPS is an all-weather, space-referenced navigation

system that allows users located anywhere on the globe or in near-Earth space to continuously and accurately determine 3D position and 3D motion and time. It belongs to the satellite navigation system, which can serve the general user for vehicle navigation, personal travel navigation, vehicle theft prevention, special groups of people to prevent loss and other purposes. Geographic information technology is now almost complete, and a large number of mobile applications and hardware devices are already using this technology.

2.2 AR of tracking registration

Tracking registration technology is the basic technology of augmented reality, and it is the key to realize the seamless fusion of virtual information and real-life scenes. There are three main types of tracking registration technology for augmented reality systems: sensor-based tracking registration technology and machine vision-based tracking registration technology and hybrid tracking registration technology. Because of the variety of hybrid tracking registration technologies, we can take the combination of machine vision and inertial sensors, the combination of machine vision, GPS and a variety of sensors depended on the needs. The tracking registration technology adopted in this system applies two tracking registration methods based on sensors and image-based natural features

2.3 Sensor technology of Android system

Today's Android phones integrate many sensors, which are tiny physical devices that can be used to obtain data such as the operating status, placement direction, ambient temperature, and magnetic field strength of mobile devices. The Android system supports multiple types of sensors, including acceleration transducer, gyroscopes, magnetic field sensors, and distance sensors. In this article, we used acceleration transducer and magnetic field sensors to get the required phone orientation information. The sensor system can automatically report sensor accuracy and data change information to the upper layer, as well as provide some interfaces for setting sensor accuracy. These interfaces can be called in JAVA applications and JAVA frameworks

2.4 OpenGL ES

OpenGL ES is a standard for mobile graphics programming. It is simplified by the OpenGL 3D graphics API standard according to the characteristics of the mobile terminal. It is a 2D/3D graphics interface designed for embedded system development. In this article, the 2.0 version is used for 3D graphics rendering. Drawing an object through OpenGL ES on an Android system depends on its three-dimensional coordinate. We can map the sensor data matrix to the OpenGLES 3D coordinate system in real time, so that the objects drawn on the OpenGLES 3D coordinate system can simulate the operation effect of the mobile phone.

3. REQUIREMENT ANALYSIS

3.1 User requirements

Our system is geared towards self-tourists, who are our target users. After the initial research of the same type of application, many famous scenic spots in China have mobile applications to introduce attractions, such as the Leshan Buddhas scenic spot application, Sanfang Seven Alleys wisdom scenic spot, but also have a collection of scenic spot guide applications such as scenic spot treasure, but the promotion is extremely poor, the number of downloads on the application platform is less than 10,000 times, the star rating is even not shown because the number of evaluations is insufficient. Without consciously searching for scenic spots on the app platform, it's difficult for visitors to see such apps promoted locally at attractions. These attraction guiding apps are not functional enough to simply integrate the search engine's tour guides into the app, lacking a truly clear guiding function.

As an AR navigation system, the service group of the system is the tourists who come to Xiamen for sightseeing. At present, we focus on the functional development of Kulangsu, a national 5A-level tourist attraction in Xiamen. When visitors go to the island, they can see the QR code on the doorplates of many old buildings. You can see the corresponding introduction by using WeChat scanning. Therefore, for the Kulangsu AR navigation system, the early building introduction is the most basic user requirement of the system. In the process of self-help travel, there is no guide to plan the scenic spots to watch the route, and it is not convenient for the tourists to plan their own routes according to the network information. At this time, it is very important to combine the route planning and navigation system in the navigation system, and it is another key user requirement of our navigation system.

3.2 Functional requirements

The functional requirements of augmented reality navigation mainly include the following aspects:

1. The system can accurately locate the geographic location information of the user's current latitude and longitude coordinates, altitude, and interest.
2. The user can select or input the target attraction, and the system will plan the walking route for the user.
3. The user can start and terminate a certain navigation at any time.
4. The system can obtain the current real scene through the mobile phone camera, and the virtual elf in the mobile phone can guide the destination to the planned route in a virtual and real way.

AR attraction introduction functional requirements include the following.

1. The user can choose the account login method to enter the system or directly enter the system without login.
2. The system can accurately correspond to the three-dimensional model, text information and animation information according to the content of the marker.
3. The three-dimensional model and information can be discarded after being identified.
4. 3D model can be scaled, rotated and other ways to interact.
5. The system can use the attraction list to select different attractions to display information.
6. Each attraction should be introduced with text, panoramic interaction or AR model interaction.

4. DESIGN OF SYSTEM

Kulangsu AR Navigation System is currently a tourism system designed for Kulangsu travel with AR navigation, scenic spots introduction and panoramic display. There are two functional modules: the first module is the Kulangsu AR navigation module, which is used for route navigation between the scenic spots on Kulangsu Island; the second module is the AR scenic spots guide module, and the AR introduction of the scenic spots can appear by identifying specific areas. You can also view the 3D virtual panorama model and related information of the scenic spots at the same time.

The core function of the AR navigation module is designed to use augmented reality technology to render the 3D route guide genie and road conditions after the user selects the target location and plans the route, and use the mobile phone sensor to make the guide genie change according to the change of the mobile phone posture, pointing to the correct direction in real time, giving users a more intuitive navigation experience.

Figure 1 is a flow chart of the Kulangsu AR Navigation System.

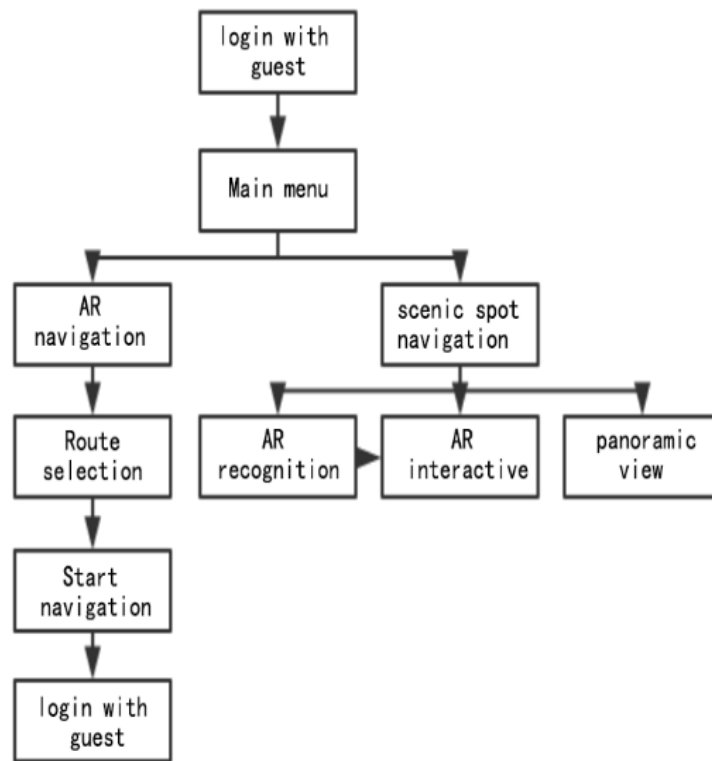


Figure 1. System flow chart

5. IMPLEMENTATION OF SYSTEM

This system uses the Unity platform and the Eclipse platform as the main development tool, and uses the GAD navigation development tool kit and the Vuforia augmented reality development toolkit to design and develop the multi-directional navigation system based on 3D model and 3D animation.

5.1 AR navigation module

In the practice process of this system, one of the most important tasks is to obtain the location information. The system is developed on the Android platform, which provides three key API for the developer to implement location positioning, namely, Location Manager, Location Provider and Location. Location Manager can be used to get the current location and track the route of the mobile device. It supports two positioning methods, including GPS positioning and network positioning. In order to obtain the user's location information better, this paper uses these two methods to locate and obtain location information.

The main difficulty of the AR navigation module lies in the visualization of navigation data. The method adopted in this paper is to convert the action identification information of the acquired road segment into a three-dimensional animation form to realize the navigation of the 3D virtual wizard tour guide in the real world. Based on this idea, this paper proposes a method to visualize navigation data.

Algorithm 1: Navigation Data Visualization Algorithm

Input: Navigation planning information IconType String Info

Output: Play animations that corresponding navigation data Animation

Method:

- 1、 Input: Navigation planning information IconType String Info
- 2、 Update()//Using the Unity function feature, each frame is called once
- 3、 String Info $\leftarrow$ jo.call(A function that returns an indication of information in a plug-in);//Ensure that new information can be obtained in time when the path is deviated from the path
- 4、 int Array[] $\leftarrow$ strToIntArray(Info);//Convert information into an integer array

Figure 2. Navigation Data Visualization Algorithm

After the data visualization is completed, the model can perform the guidance combined virtuality and reality before acquiring the image of the camera according to the obtained navigation instruction information. Click the Destination button on the page to enable navigation from the current location to the destination. The test results are as follows:

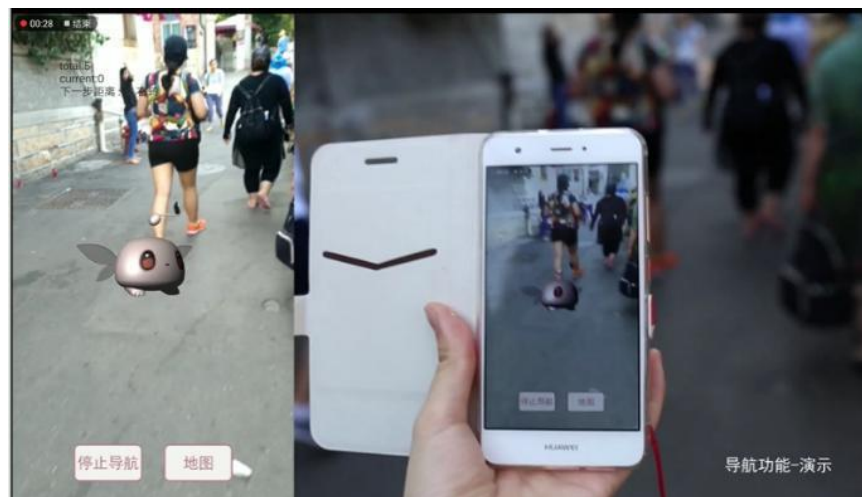


Figure 3. Navigation test

It can be seen from Fig. 2 that the system successfully obtains the number of road segments from the starting point to the target location, the current road segment where the simulated user is located, the length to the next road segment and the guide action of the next segment.

5.2 AR viewpoint guide module

1. data input: collect video flow through camera and save it as image data; gray processing of colour image in order to improve the efficiency and accuracy of subsequent detection.
2. tracking registration: calibrating the position and posture of the camera, obtaining the camera parameters, using the SURF algorithm to detect the feature points in the image data, matching the feature data in the database, and calculating the homography matrix through the camera parameters and matching feature points, so as to gain the pose data of the virtual camera.
3. Combination of The Actual and Visual Conditions: the posture data of virtual camera, the fusion of image data and virtual resources, rendering output.

Figure 4 and Figure 5 are the test diagrams of the guide module.



Figure 4. Guide test 1



Figure 5. Guide test 2

As can be seen from Figure 3 and 4, scanning the door card through a mobile camera can successfully transfer the detailed introduction of the viewpoint, as well as the virtual and real combination of the 3D model and the reality model, and realize the interaction.

6. CONCLUSION

It is a valuable research direction to combine augmented reality technology with tourism navigation. This system uses the Unity3D platform for the overall development on the Android development environment, uses the Vuforia toolkit to develop the augmented reality technology module, and combines the geographic information service technology and the panorama technology. It combines the two types of tracking and registration based on the sensor and the natural features of the image, and studies and implements a famous viewpoint on Kulangsu, Xiamen as the background of tourism navigation and AR scenic spot introduction system.

After testing, the navigation module can obtain the positioning navigation information on the mobile devices, and realize the real-time navigation of the 3D virtual tour guide into the real world scene. Viewspot navigation module can receive the relevant information of the scenic spot through the AR application on the mobile phone, view the 3D virtual scenic spot model, and make a simple interaction.

The system has initially completed the development of the basic functions of augmented reality navigation, allowing users to take a more visual tour on Gulangyu Island through the model animation in the screen. However, the current navigation module of the system still has areas for improvement.

1. the current three-dimensional model used for navigation is temporarily unable to identify the ground and create on the ground, the use of mobile phone gyroscopes in the virtual space orientation determination method is still defective, not yet used.
2. the three-dimensional model used for navigation can only be in front of the mobile phone lens and the user is a fixed distance away from the model that has not yet reached the desired goal has its own line of action.
3. the fusion of reality and reality is still not enough, the light mapping of the model still needs a complex implementation method, so as to distinguish the real world lighting situation in real time, and apply it to the rendering of the model.

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